

What a Chance Constraint Does Not Promise

Measuring a demand-ceiling guarantee across thirty-seven electricity supplies in eight countries, with India as the focal case

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Abstract

Indian commercial demand charges are set by a single worst thirty-minute block in a month, which makes defending a demand ceiling a risk problem rather than a prediction problem. The standard applied response is to substitute a high forecast quantile into the capacity constraint and call the result a 95% guarantee. We measure what that guarantee is actually worth, on fourteen held-out windows across an American office and an Indian campus at native 15-minute resolution. The per-step exceedance rate is 0.03–0.08 against a nominal 0.05 on every window — marginally, the forecaster does what it promised — while the probability of breaching the ceiling *somewhere* in the controller’s 16-hour planning window is 0.39 to 0.82, median 0.57: eight to sixteen times the level the constraint appears to promise, and not a constant, since one meter returns 0.39, 0.64 and 0.42 in three successive Junes while its per-step rate does not move. The gap narrows but persists — median nine times, lowest day-block bootstrap bound three times — when the per-step layer’s adaptive step is restated in units that transfer across meter sizes. Neither textbook substitute recovers it: the independence calculation $1 - (1 - \alpha)^H$ overstates the risk by $1.7\times$ at the median because load errors are strongly autocorrelated, and the Boole/Bonferroni union bound — the only one of the two that is a valid bound — is 1 on every window. A Student- t copula over the horizon predicts the realised value to a mean absolute error of 0.065; we use it to restore the level as a settable parameter with a scenario MILP, tracking requested ε at rank correlation 0.957 for Rs 3,339 on a billing month. On a second control object — an Indian campus feed under Delhi’s own tariff — the response stays monotone (rank correlation 0.988) and the exactness does not survive, on the one series here whose per-step layer under-covers — and repairing that layer moves the dial the right way by an amount one billing month cannot resolve. We report its exactness and resolution limits.

We test the rest of the pipeline by running one unchanged protocol over thirty electricity supplies in eight countries: eighteen buildings across eight usage classes from the Building Data Genome Project 2, six metered national or city demand series including the Delhi State Load Despatch Centre and ENTSO-E, and six Chinese provinces added late on an externally imposed constraint. Two findings follow. Distribution-free coverage appeared to degrade with aggregation level — 0.901 on buildings against 0.829 on metered system demand — and does not: the adaptive layer’s step is a number in the units of the series, and at system scale it did not move. Under split conformal alone the tiers are within 0.03 of each other and eleven of eighteen buildings miss 0.85, so the exchangeability failure is universal; with the step stated as a fraction of the interval width the same layer holds 0.90–0.93 at every tier and no supply in the study misses 0.85. And forecast skill is uncorrelated with climate while available flexibility tracks it strongly, so deployment screening should ask how much load is weather-driven rather than how predictable it is; the null replicates on the Chinese panel it was not fitted to.

India is the focal case and returns a divided verdict: the most favourable load physics in the study and the worst calibration in it. The China arm’s most quotable result — that the closest structural analogue to Delhi returns the second-worst skill in the benchmark — is retracted in Section 9, because auditing that dataset against its own documentation showed a whole year of provincial demand to be 778 numbers reproducing 8,760 values to the precision the source is published at, which makes its skill column a statement about the data rather than about a forecaster. Five rows argue against the method, three of them outright negative, and all five are reported at the same size as the rest.

1 Introduction

An Indian commercial electricity bill has three levers: a time-of-day energy charge, a demand charge levied on the highest 30-minute average demand in the month, and power-factor penalties. The second dominates the risk. It is set by one block out of roughly 1,440 in a month, so a single bad afternoon

prices all thirty days. A controller that reduces average consumption but occasionally overshoots can cost more than one that does nothing.

The standard applied response is a chance constraint: replace the forecast mean in the capacity constraint with a high quantile, typically the 95th, and treat the result as a 95% guarantee. Formally, with b_t the uncontrollable base load, u_t the controllable allocation, s_t solar generation and D the ceiling,

$$\hat{b}_t^{q95} + u_t - \hat{s}_t^{q05} \leq D \quad \text{for each } t \text{ in the horizon.} \quad (1)$$

Equation (1) is a set of *marginal* statements. Each row says that at that timestep, considered alone, the realisation exceeds the bound about 5% of the time. The event that costs money is different: it is

$$\Pr \left[\max_{t \in \mathcal{W}} (b_t + u_t - s_t) > D \right], \quad (2)$$

a statement about a maximum over a window $\mathcal{W}$. That (1) does not imply (2) is elementary and well known; joint chance constraints, their intractability, and the standard routes around them — Boole/Bonferroni risk allocation and the scenario approach [Calafiore and Campi, 2006, Nemirovski and Shapiro, 2006] — are textbook material.

What is not established is how large the gap is on real building load, and how badly each of the standard substitutes misses it. Both ends matter practically: the first is how much a deployed controller is under-protected, and the second is how much headroom a careful designer gives away for nothing. We measure both, and find they are wrong in opposite directions by factors we can state.

Contributions.

1. A measured bracket on horizon risk for a demand ceiling on held-out meter data (Section 5), over fourteen windows in two countries: claimed 0.05, realised 0.39–0.82 (median 0.57; no 95% lower bound under 0.28), independence calculation 0.87–0.99, union bound vacuous on every row, and the gap surviving a restatement of the per-step layer’s own tuning parameter in units that transfer across meter sizes. The common formulation is under-protected by a factor of eight to sixteen, the usual correction is $1.7\times$ too conservative at the median, and the gap is a distribution rather than a number.
2. A closed-loop demonstration, on two control objects in two countries under their own tariffs, that this risk level can be made an operator-set parameter rather than an emergent property of horizon length — with the half of that claim which replicates separated from the half which does not. The response is monotone on both (rank correlation 0.957 and 0.988), and a matched-pair test resolves it step by step within one billing month; it is close to exact and errs safe only on the plant whose per-step layer is itself calibrated, and repairing that layer on the other moves the dial the right way by an amount one billing month cannot resolve (Section 6).
3. A finding made and then unmade by its own audit. One protocol over thirty supplies in eight countries, replicated inside a single city across building, campus and system demand, showed conformal coverage degrading with aggregation. The audit of the adaptive layer at system scale showed why: its step is stated in the units of the series and had stopped moving. Re-read without that confound, split conformal fails at every tier alike and the adaptive layer, with its step as a fraction of the interval width, holds the level at every tier (Section 7). What transfers across scales is not the number but the fraction, and a practitioner moving the layer from a building to a feeder will hit exactly this.
4. A null separating forecast skill from available flexibility as independent deployment axes, replicated on a panel it was not fitted to (Section 8).
5. A provenance audit of a dataset added to the study under an external constraint, which disqualifies that arm’s most favourable result and is reported as such (Section 9).

Where the novelty is, and where it is not. We invented no method. Every component — chance-constrained MPC for buildings, joint chance constraints and their standard treatments, quantile gradient boosting, conformalized quantile regression, adaptive conformal inference, copula scenario generation, decision-focused learning — is prior work, and Section 2 places each. The novelty is empirical and integrative: an instrument that pins everything and moves one thing, run on metered load, with the results that argue against it reported at full size. Section 2 states the gap that survives the literature.

2 Related work, and the gap

Four literatures meet at a demand ceiling, and the gap sits where they fail to overlap.

Stochastic and chance-constrained MPC for buildings. Oldewurtel et al. [2012] and Oldewurtel et al. [2014] established chance-constrained MPC for building climate control under weather uncertainty, reformulating per-step chance constraints deterministically under Gaussian disturbance assumptions; Zhang et al. [2013] replaced the distributional assumption with sampled scenarios in the sense of Calafiore and Campi [2006]. Joint-in-time chance constraints — that a trajectory stay feasible at *every* step of the horizon — are handled in the general stochastic-MPC literature by risk allocation under Boole’s inequality [Ono and Williams, 2008, Paulson et al., 2020], which is conservative by construction and, as Section 5 measures, vacuous at the horizon a demand ceiling requires. A recent line combines conformal prediction with MPC to obtain joint-in-time regions without a distributional assumption [Lindemann et al., 2023, Vogel et al., 2025], evaluated on trajectory-prediction and synthetic linear-system benchmarks. The one application of that idea to building demand response we are aware of [Xu et al., 2026] uses stage-wise — that is, marginal — conformal tightening in an EnergyPlus simulation and does not report horizon-level violation. None of these measures, on metered building load, how far a marginally calibrated q_{95} constraint’s realised horizon risk sits from its nominal level.

Demand-charge management. The closest prior work on monthly peak limiting under load uncertainty is the Hong Kong PolyU line [Xu et al., 2019, 2020]: probabilistic ANN load forecasts, driven by probabilistic weather forecasts, feed an expected-cost threshold resetting scheme, and Monte Carlo is used to study the robustness of the *threshold*, not to bound the probability of exceeding it. Flamm et al. [2020] treat demand-charge management with storage as multi-period stochastic MPC over weather scenarios with affine decision rules and an expected-cost objective, with a weighting for horizons that straddle billing periods. In neither line is the exceedance probability of the ceiling a constraint with a stated level, and neither is evaluated on an Indian tariff or an Indian building. Field and simulation studies of HVAC demand flexibility under MPC report peak reduction and savings without a risk statement at all.

Conformal prediction for multi-step forecasts. Split conformal [Vovk et al., 2005] and CQR [Romano et al., 2019] give per-step marginal coverage. Stankevičiūtė et al. [2021] obtain joint coverage over a horizon by Bonferroni correction; Sun and Yu [2024] replace Bonferroni with a copula over per-step conformity scores, which is the same modelling move as Section 6 but deployed as a *predictor* rather than inside an optimiser, and requiring a second calibration set. Lin et al. [2022] study conformal intervals under temporal dependence. Zaffran et al. [2022] and Wang and Hyndman [2024] extend adaptive conformal inference [Gibbs and Candès, 2021] to non-stationary and multi-step settings, targeting marginal coverage; the latter establishes that optimal h -step errors are serially correlated to lag $h - 1$, which is the dependence our copula fits. This paper adds to that line a measurement rather than a method: the realised joint exceedance on real load errors, placed against both the Bonferroni bound and the independence figure, with the copula’s accuracy for that quantity reported.

Decision-aware forecasting and calibration. Decision-focused learning [Elmachtoub and Grigas, 2022, Donti et al., 2017, Berthet et al., 2020] trains the forecast on the downstream decision’s cost; Stratigakos et al. [2026] calibrate multivariate prediction sets to the *reliability of the downstream decision* rather than to coverage, for robust reserve scheduling. That is complementary to Section 6, which reaches the same object — an operator-set violation level on a committed ceiling — from the optimiser side, and reports its exactness and resolution floor.

Data. Probabilistic load forecasting is mature at system scale [Hong and Fan, 2016, Hong et al., 2016] and increasingly benchmarked at building scale [Miller et al., 2020, Emami et al., 2023, Nweye et al., 2023]. Aggregation is known to improve point accuracy by a scaling law [Sevlian and Rajagopal, 2018]; whether *distribution-free coverage* holds across aggregation levels has not, to our knowledge, been reported. For India, Rashid et al. [2019] publish I-BLEND: 52 months of one-minute mains data, with power factor, for seven buildings on a Delhi campus. It is the only open Indian building-level load corpus at the resolution Indian demand charges are assessed on, and it has been used for point forecasting rather than for demand-charge control.

The gap. We find no published work that (i) measures, on metered building load, the horizon-level breach probability a marginally calibrated q_{95} chance constraint actually delivers against its nominal level, bracketed by the valid Boole bound above and the independence figure that is usually misquoted as one; (ii) restores that level as a settable parameter inside a demand-charge MILP and reports its exactness and resolution floor; (iii) tests distribution-free coverage across building and system scale under one protocol; or (iv) does any of this on real Indian commercial buildings at native 15-minute resolution under an Indian tariff. Items (i)–(iii) are this paper. Item (iv) is the work Section 11 opens, and the data for it exists.

3 The instrument

Six components, each independently swappable, with the bill engine as the single source of truth for every reported figure:

tariff $\rightarrow$ quantile forecast
 $\rightarrow$ conformal calibration
 $\rightarrow$ chance-constrained MILP
 $\rightarrow$ RC simulator $\rightarrow$ bill engine.

The forecaster is a quantile LightGBM [Ke et al., 2017] at five levels over a 64-step (16-hour) horizon at 15-minute resolution, wrapped in split and adaptive conformal calibration [Angelopoulos and Bates, 2023, Gibbs and Candès, 2021]. The controller is a deterministic MILP solved with HiGHS; no language model touches a money decision. The evaluation discipline is that a number reaches a table only if a script produced it, and a single entry point regenerates every figure and every table in this paper from a clean checkout.

This matters for the measurements because each one requires pinning everything and moving one thing. Section 5 pins the plant and varies only the risk formulation. Section 7 pins the protocol and varies only the series. Section 8 pins the model and varies the site. Section 9 pins all three and varies the country, which is the one we did not choose.

4 Data

Six public sources, all fetched by script rather than vendored, and named in full because the benchmark’s credibility rests on them.

Building Data Genome Project 2 [Miller et al., 2020]. 1,636 metered non-residential buildings across nineteen sites, of which 1,578 carry an electricity meter, hourly with matched weather, two years. It is the only harmonised multi-country building corpus in existence, which is why it carries the controlled comparison: the same physical quantity, the same metering convention, the same period. It contains *no Indian buildings*, which is why the next source is in the study.

I-BLEND [Rashid et al., 2019]. One-minute mains power and power factor for seven buildings on the IIT-Delhi campus — academic, lecture, library, facilities, dining and two dormitories — and for the three transformers that feed the whole campus, from August 2013 to December 2017, CC0. The billing blocks are formed by averaging sixty one-minute readings, not by interpolating up from hourly data, so this is the only building-level source in the study at the cadence Indian demand charges are assessed on. Its documentation states, and the load–temperature signature of each meter confirms, that every building except Facilities is served by a central chiller plant that is *not* on the building’s own meter: at building level the corpus is therefore a forecasting and calibration benchmark rather than a control one, and the selection rule’s chilled-water clause records it as such. The campus feed is the exception, and it is the more important object: an Indian campus is billed as a single high-tension consumer with one contract demand and one monthly demand charge, so the transformer total — median 199 kW, peak 842 kW, chiller included — is what the tariff actually sees. Meter uptime varies by feed; each series’ admissible test Junes and the training months behind each are recorded by `data/iblend.py` and printed in the table.

H	hours	per-step $\hat{\alpha}$	realised	95% CI	independent	Boole $H\hat{\alpha}$	copula
1	0.25	0.060	0.060	[0.03, 0.09]	0.060	0.060	0.049
4	1.00	0.060	0.091	[0.05, 0.13]	0.219	0.240	0.100
8	2.00	0.060	0.126	[0.08, 0.18]	0.389	0.478	0.148
16	4.00	0.059	0.182	[0.11, 0.25]	0.625	0.952	0.225
32	8.00	0.061	0.275	[0.18, 0.38]	0.867	1.000	0.342
64	16.00	0.062	0.405	[0.28, 0.53]	0.983	1.000	0.497

Table 1: Marginal versus horizon-level exceedance of the demand ceiling, held-out June 2017, 2,817 forecast origins over 31 days. The per-step rate $\hat{\alpha}$ is well calibrated at every horizon against a nominal 0.05. The realised probability of breaching *somewhere* in the window is not that number: at $H = 64$ it is between 6 and 11 times the nominal level (day-block bootstrap, 95% interval; adjacent origins share 63 of 64 steps, so the day is the unit of replication). Neither substitute is usable: the independence calculation $1 - (1 - \hat{\alpha})^H$ overstates the risk by $2.4\times$ because load errors are strongly autocorrelated, and the Boole/Bonferroni union bound $\min(1, H\hat{\alpha})$ – the only one of the two that is a valid bound – saturates at 1 by $H = 17$ and is vacuous over the controller’s actual horizon.

Delhi State Load Despatch Centre. Fifteen-minute metered demand for the city of Delhi, disaggregated across five distribution companies (BRPL, BYPL, NDPL, NDMC and MES) plus the total, two complete years at full coverage. With I-BLEND this gives three levels of aggregation — building, campus, city — in one city, at native resolution, under one weather feed. Every non-Indian series in the study is interpolated up from hourly.

ENTSO-E transmission data, via Open Power System Data. Hourly national load for the United Kingdom, Ireland, Germany, France and Spain, drawn from the European transmission system operators’ transparency platform.

Chinese provincial demand [Wu and Kan, 2023]. Hourly load for the 31 provinces of mainland China, 2018, from which six are drawn. This source was added after the study was otherwise complete, and it is the only tier-2 series that is constructed rather than metered. Section 9 audits it.

ERA5 reanalysis, via the Open-Meteo archive. Hourly temperature, dew point and cloud cover at each location, used identically for every series.

Windows. The building tier and the European national tier share 2016–2017. The Delhi archive covers 2011–2012, so the same protocol is shifted back five years with its shape preserved. The Chinese series is a single calendar year, 2018. In every row of the study the test month is June, and the headline metric is skill against each series’ own seasonal-naïve baseline: a ratio computed inside one series’ own data and its own period, which is what makes rows with different units and different years comparable at all. Absolute losses are not comparable across groups and are never compared.

Selection. One rule, stated in advance and applied uniformly: no district chilled-water meter, $>97\%$ coverage, median load 40–800 kW, positive load–temperature correlation; ties within a (site, usage) cell broken by highest $p99/\text{median}$. Thirty supplies pass. Two facts about the rule were discovered by applying it rather than anticipated when writing it, and both are reported in Section 7.

5 The horizon gap

Table 1 and Figure 1 report, for a 13,759 m² office on held-out June 2017 data over 2,817 forecast origins, the per-step exceedance rate of the ceiling alongside the probability that the ceiling is breached at least once within a window of H steps.

Three readings of the last row are simultaneously true and together are the result.

The forecaster is not the problem. The per-step quantile is well calibrated at every horizon: $\hat{\alpha}$ holds between 0.059 and 0.062 against a nominal 0.05. Marginal calibration delivers exactly what it promised, which is why this is a result about a formulation rather than a bug report about a model.

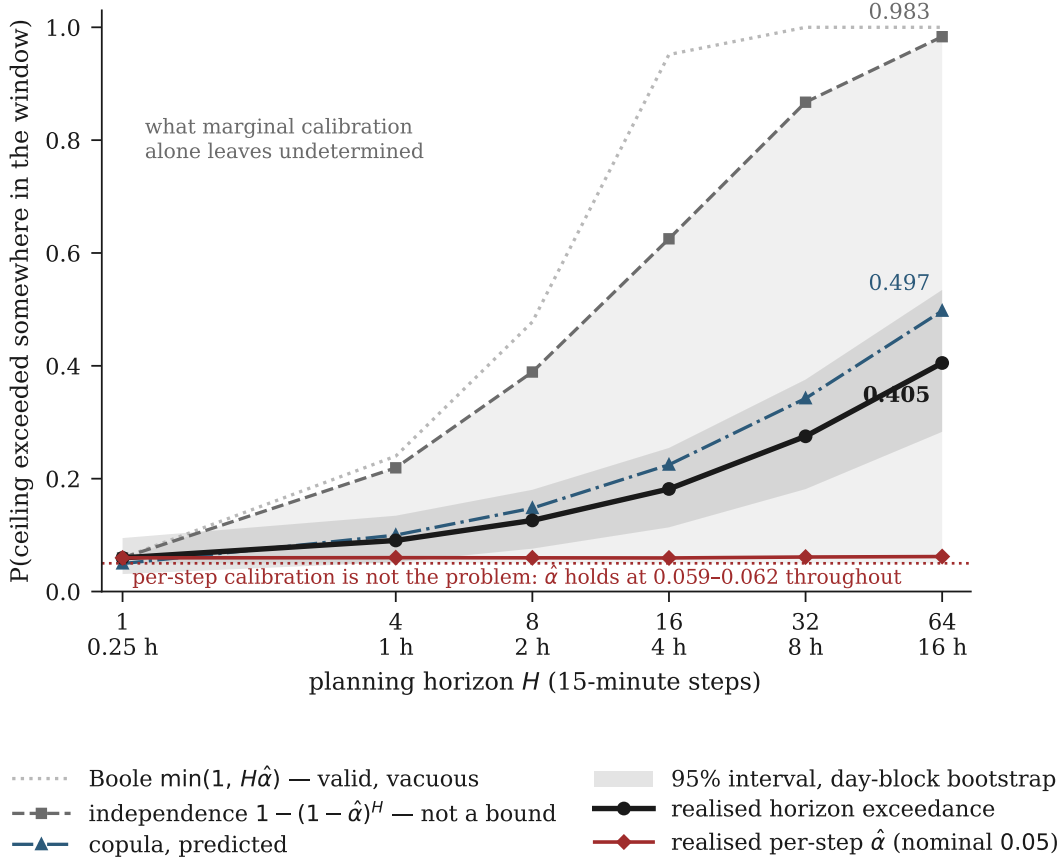


Figure 1: The bracket. Marginal calibration is a statement about each row of (1) separately; it constrains the probability of breaching somewhere in the window only to the shaded region, which at $H = 64$ spans 0.05 to 0.983. The per-step rate (bottom) is flat and well calibrated at every horizon, so nothing here is a broken forecaster. The band around the realised curve is the 95% day-block bootstrap interval. Held-out June 2017, one 13,759 m² office, 2,817 forecast origins.

The formulation is under-protected by a factor of eight. The probability of breaching somewhere in the 16-hour window is 0.405: $6.5\times$ the per-step rate the forecaster actually delivers, and $8.1\times$ the 0.05 that (1) appears to promise. That figure carries sampling error, and the naive kind would understate it: adjacent origins share 63 of their 64 steps, so the unit of replication is the day rather than the origin, and a day-block bootstrap puts the 95% interval at 0.28–0.53, between $5.7\times$ and $10.7\times$ nominal. A controller deployed on (1) and described to an operator as a 95% guarantee is running at roughly 60%, and at no better than 72%.

Neither standard substitute recovers it, and they fail in opposite directions. This is the correction to the usual telling, and it matters. The quantity commonly quoted, $1 - (1 - \alpha)^H$, is the *independence* calculation, not a bound: it assumes the exceedance events are independent, which is the one thing load errors are not. It gives 0.983, which is $2.4\times$ the realised 0.405, because the fitted error process is heavily autocorrelated — lag-1 correlation 0.871, 0.578 at lag 4, decaying to 0.085 by lag 32 — so exceedances cluster rather than spread. The quantity that *is* a valid bound, Boole/Bonferroni’s $\min(1, H\hat{\alpha})$, makes no dependence assumption and pays for it: at $\hat{\alpha} \approx 0.06$ it reaches 1 by $H = 17$, four hours into a sixteen-hour horizon, after which it says nothing at all.

The practical consequence runs in both directions. A controller using (1) is materially under-protected. A controller sized off the independence figure is over-protected by more than a factor of two and pays for it in usable headroom, which is the quantity the entire economic argument rests on. A controller that insists on a valid bound over this horizon has no bound to use.

The bracket replicates, and it is a distribution. The measurement above is one building in one month. Table 2 and Figure 2 repeat it, with nothing changed but the series, over thirteen windows on the IIIT-Delhi campus of Section 4 — six buildings and the campus feed, up to three Junes each,

Series	June	per-step $\hat{\alpha}$	realised $H=64$	95% CI	independent	Boole	copula	lag-1 ρ
Phoenix office (BDG2)	2017	0.062	0.405	[0.28, 0.53]	0.983	1.00	0.497	0.87
IIIT-Delhi Academic	2015	0.049	0.393	[0.28, 0.50]	0.959	1.00	0.477	0.77
IIIT-Delhi Academic	2016	0.050	0.636	[0.50, 0.75]	0.963	1.00	0.580	0.72
IIIT-Delhi Academic	2017	0.048	0.422	[0.30, 0.54]	0.956	1.00	0.595	0.71
IIIT-Delhi Boys	2014	0.032	0.501	[0.38, 0.61]	0.874	1.00	0.554	0.72
IIIT-Delhi Campus	2017	0.079	0.432	[0.29, 0.58]	0.995	1.00	0.482	0.85
IIIT-Delhi Dining	2017	0.053	0.714	[0.61, 0.81]	0.970	1.00	0.744	0.72
IIIT-Delhi Facilities	2015	0.052	0.639	[0.55, 0.71]	0.967	1.00	0.705	0.63
IIIT-Delhi Facilities	2016	0.054	0.621	[0.52, 0.72]	0.972	1.00	0.633	0.66
IIIT-Delhi Facilities	2017	0.048	0.739	[0.62, 0.85]	0.958	1.00	0.753	0.61
IIIT-Delhi Girls	2014	0.048	0.693	[0.59, 0.78]	0.957	1.00	0.589	0.63
IIIT-Delhi Girls	2017	0.046	0.818	[0.74, 0.89]	0.951	1.00	0.739	0.46
IIIT-Delhi Library	2014	0.036	0.480	[0.36, 0.59]	0.902	1.00	0.567	0.73
IIIT-Delhi Library	2016	0.042	0.509	[0.41, 0.61]	0.935	1.00	0.504	0.80

Table 2: The bracket of Table 1, replicated over 14 windows: the Phoenix office of the original measurement and 13 Indian windows at native 15-minute resolution. The per-step rate is calibrated throughout (0.032–0.079 against a nominal 0.05). The realised probability of breaching somewhere in the 16-hour window runs from 0.393 to 0.818, median 0.565 — between 8 and 16 times the level the constraint appears to promise. Intervals are day-block bootstrap, the day being the unit of replication because adjacent origins share 63 of 64 steps. No window’s lower bound falls under 0.28, $5.7\times$ nominal. The spread across windows is not sampling noise: 32 of 91 pairs of intervals are disjoint, the lowest and highest rows among them, and the same building returns different values in different years while its per-step rate does not move. A 16-hour window opened late in one day runs into the next, so two-day blocks are also reported: they widen the intervals by 13%, and the floor becomes 0.25 ($5.0\times$) with 20 pairs disjoint. The independence figure overstates realised risk by $1.66\times$ (median) and lies inside the realised interval on 0 of 14 windows; the Boole bound is 1 on every row. The copula predicts the realised value to a mean absolute error of 0.065 against 0.382 for independence, and lies inside the interval on 12 of 14 windows.

every one at native 15-minute resolution rather than interpolated from hourly. Three things hold on every row. The per-step rate is calibrated, 0.032–0.079 against a nominal 0.05. The Boole bound is 1. And the independence figure sits between 0.87 and 0.99, overstating the realised risk by $1.66\times$ at the median. What does not hold is the number itself. The realised probability of breaching somewhere in the window runs from 0.393 to 0.818, median 0.565 — between eight and sixteen times the level the constraint appears to promise — and it is not a constant of the formulation: the Academic building returns 0.393, 0.636 and 0.422 in three successive Junes while its per-step rate stays within 0.002 of 0.05. The Phoenix figure of 0.405 was at the mild end of the range. The intervals say the spread is not sampling noise: no window’s lower bound falls under 0.28, $5.7\times$ nominal, and 32 of the 91 pairs of intervals are disjoint, the lowest and highest rows among them. Two-day blocks, which allow for a window opened late in one day running into the next, widen the intervals by 13% and leave 20 pairs disjoint. The gap is a property of how exceedances cluster in a given month, which marginal calibration says nothing about, and the practical reading is that an operator cannot even be told a single corrected number: the correction is itself a random variable with a wide law.

The copula is the one estimate that tracks it. Across the fourteen windows it predicts the realised value to a mean absolute error of 0.065, against 0.382 for the independence figure, and lies inside the realised value’s 95% interval on twelve of the fourteen; the independence figure lies inside none. It is not exact — it misses the two most extreme rows by 0.11 and 0.13 — and Section 6 states its limits. But it is the only quantity in the table that moves with the truth, and that is the property the dial in the next section depends on.

The gap is not an artefact of how the per-step layer was tuned, but its size is not independent of it either. Every row above was calibrated with an adaptive layer whose step is a number in the units of its own series, and these meters run from intervals of five kilowatts to two hundred and twenty. Section 7 shows what that did to the calibration table; the same objection applies here, because a per-step rate is what the first column reports. Table 3 repeats all fourteen windows with the step restated as a fraction of the interval width, $\kappa = 0.006$, the width read on the validation block. Two things follow, and only one of them was expected. The gap survives: realised 16-hour exceedance runs 0.254 to 0.660, median 0.440, nine times the level the constraint appears to promise, and the lowest bound anywhere in the table is 0.15, three times nominal. But the gap also shrinks, by 0.111 on average, and on six of the fourteen windows the two measurements are separated by more than sampling error. The mechanism is

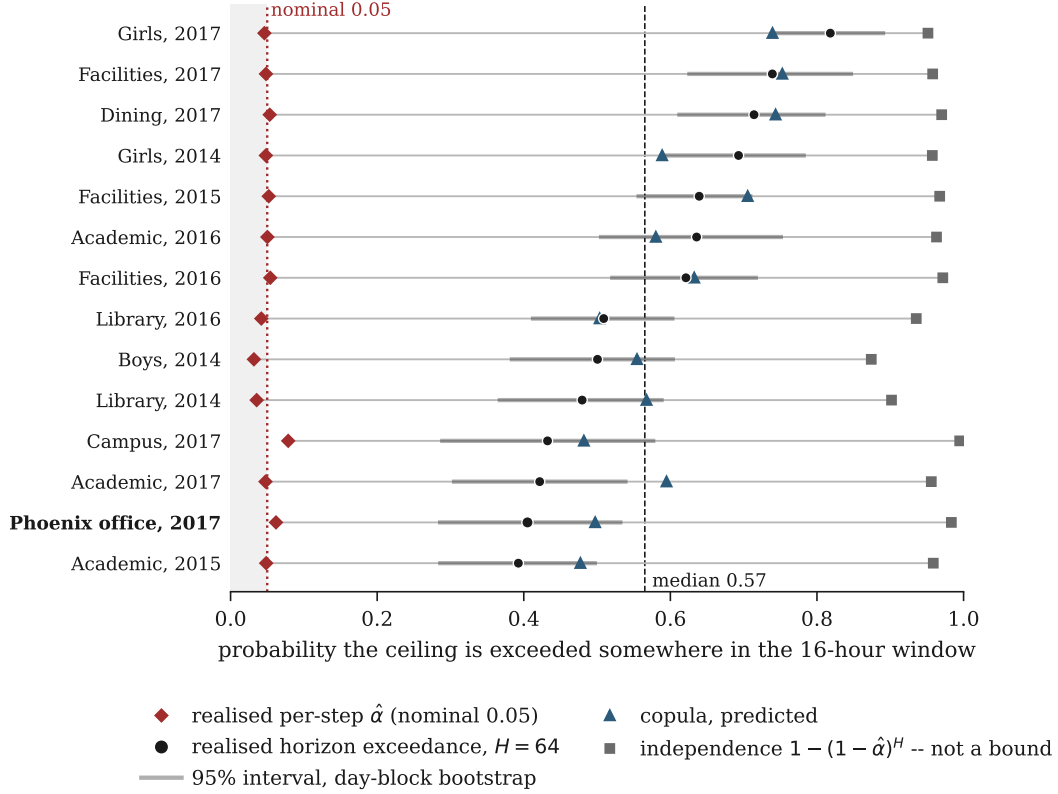


Figure 2: The bracket, one row per window, sorted by the realised value. The per-step rate sits on the nominal line on every row; the realised horizon exceedance does not sit anywhere in particular, and the same meter returns different values in different Junes. The independence figure is near 1 throughout. Phoenix is the original measurement; the rest are IIIT-Delhi at native 15-minute resolution.

visible in the first two columns: on these small meters the absolute step was $\kappa = 0.02$ to $\kappa = 0.07$, a *faster* layer than the relative one and not a slower one, and a fast layer produces a bound that jumps after each exceedance and decays between them. That bound is noisier, its exceedances are closer to independent across the horizon, and the probability that at least one occurs is correspondingly higher. A slower layer gives a smoother bound, more clustered exceedances, and a smaller joint probability — at the cost of tracking the nominal per-step rate less tightly, which is why the panel above, whose per-step column is the better calibrated of the two (0.0073 against 0.0144 mean absolute distance from nominal), is the one reported as the measurement.

The honest statement is therefore narrower than “the gap is 8 to 16 times” and survives being narrowed. Under both calibrations of the per-step layer, on every one of fourteen windows in two countries, a marginally calibrated q_{95} constraint delivers a horizon-level breach probability between three and sixteen times its nominal level, and which end of that range a particular deployment lands on depends on a tuning choice nobody making it would expect to be choosing a risk level.

6 Restoring the guarantee as a dial

We fit a Student- t copula over the horizon and sample joint trajectories — the marginals-to-scenarios construction of Pinson et al. [2009], with a heavier-tailed dependence family — which a scenario MILP then constrains with a CVaR surrogate [Rockafellar and Uryasev, 2000, Nemirovski and Shapiro, 2006] on the probability mass permitted to violate. The copula reproduces the calibrated marginals to within 4.6% of mean interval width at 4,000 paths per origin, so it adds dependence and disturbs nothing upstream, and it predicts horizon exceedance far closer than either substitute: 0.497 against a realised 0.405 (interval 0.28–0.53) at $H = 64$, where (1) implies 0.05 and the independence figure is 0.983.

Table 4 is a closed-loop run over a real billing month at two demand targets and four risk levels. The acceptance metric is violation of the ceiling *the optimiser itself committed to at each origin*, because that is the event ε is a statement about. The operator’s monthly demand target is a business metric

Series	June	$\hat{\alpha}$ abs.	$\hat{\alpha}$ rel.	realised abs.	realised rel.	95% CI	copula
Phoenix office (BDG2)	2017	0.061	0.072	0.426	0.425	[0.29, 0.56]	0.452
IIIT-Delhi Academic	2015	0.049	0.047	0.393	0.342	[0.24, 0.45]	0.428
IIIT-Delhi Academic	2016	0.050	0.055	0.636	0.572	[0.43, 0.71]	0.481
IIIT-Delhi Academic	2017	0.048	0.049	0.422	0.351	[0.22, 0.49]	0.441
IIIT-Delhi Boys	2014	0.032	0.025	0.501	0.325	[0.22, 0.43]	0.420
IIIT-Delhi Campus	2017	0.079	0.069	0.432	0.438	[0.30, 0.57]	0.528
IIIT-Delhi Dining	2017	0.053	0.054	0.714	0.660	[0.55, 0.76]	0.656
IIIT-Delhi Facilities	2015	0.052	0.074	0.639	0.603	[0.51, 0.69]	0.602
IIIT-Delhi Facilities	2016	0.054	0.082	0.621	0.609	[0.49, 0.72]	0.546
IIIT-Delhi Facilities	2017	0.048	0.053	0.739	0.516	[0.37, 0.65]	0.614
IIIT-Delhi Girls	2014	0.048	0.034	0.693	0.442	[0.31, 0.58]	0.378
IIIT-Delhi Girls	2017	0.046	0.042	0.818	0.632	[0.51, 0.75]	0.450
IIIT-Delhi Library	2014	0.036	0.033	0.480	0.254	[0.15, 0.37]	0.399
IIIT-Delhi Library	2016	0.042	0.026	0.509	0.307	[0.20, 0.40]	0.378

Table 3: The horizon rows of Table 2 with the per-step layer’s adaptive step stated as a fraction of the interval width ($\kappa = 0.006$, the width read on the validation block) instead of as an absolute number in the units of each series. These meters run from intervals of a few kilowatts to several hundred, so the absolute step was a different layer on every row — $\kappa = 0.0015$ to $\kappa = 0.0691$ across these windows, so on all but the campus feed it was faster than the step the building audit settled on, not slower. Under the relative step the per-step rate spans 0.025–0.082 against 0.032–0.079 before, and its mean absolute distance from the nominal 0.05 moves from 0.0073 to 0.0144: a slower layer tracks the nominal rate less tightly, which is why the per-step column of Table 2 is the one reported there. The horizon gap survives and shrinks. Realised 16-hour exceedance runs 0.254–0.660 (median 0.440, $9\times$ nominal) against 0.393–0.818 (median 0.565) before; the paired change is -0.111 on average, and the absolute-step value lies inside the relative-step row’s own 95% interval on 8 of 14 windows, so on 6 the two are separated by more than sampling error. The magnitude of the gap is therefore not independent of how the per-step layer is tuned — a faster layer produces a noisier bound and more nearly independent exceedances across the horizon — but no choice of step makes it go away: the lowest lower bound anywhere in this table is 0.15, $3.0\times$ the level the constraint appears to promise. The copula lies inside the realised interval on 12 of 14 windows. Fox is run through this trainer at both steps, because its row in Table 2 comes from the benchmark harness and would not otherwise be a paired comparison.

the chance constraint never promised: the committed peak is a decision variable, and an optimiser short of capacity may lift it, pay the demand charge, and honour its constraint exactly. Grading a chance constraint against a number it never promised is the usual way this test is made meaningless, so both columns are shown and only one is the acceptance criterion.

The result is that ε becomes a dial. Realised commit violation tracks the requested level with rank correlation 0.957 across the sweep, mean absolute gap 0.057, and lands conservative at five of eight levels. The formulation errs safe by construction: the CVaR surrogate admits a strict subset of the chance-constrained feasible set, so it defends the committed ceiling harder than asked. Those are point estimates over one month, and the commitments behind them are overlapping sixteen-hour windows, so each rate carries a day-block interval; the mean half-width is 0.081. The requested ε lies inside its own interval at six of eight levels, and the two it does not are both $\varepsilon = 0.35$, on the safe side: on this plant the dial is never measurably unsafe.

A second control object, and the half of this that replicates. Table 5 repeats the sweep on the IIIT-Delhi campus feed — another country, another tariff, another plant, and the consumer a Delhi utility actually bills — with nothing tuned for it. Two things happen, and only one of them is the result above.

The dial’s *monotonicity* replicates, and more cleanly than on the office: rank correlation 0.988, and realised commit violation rises through 0.172, 0.236, 0.300, 0.392 as ε is set to 0.05, 0.10, 0.20, 0.35. An operator turning the dial gets a monotone response on both plants.

Whether one month *resolves* that response is a different question from whether each level’s rate is exact, and it has a sharper answer than the intervals in the tables suggest. Those intervals are wider than the spacing between adjacent levels, but they are the wrong instrument for comparing levels: every level at a target is tested from the same states, so the commitments can be matched origin by origin. Matched, the outcomes are almost nested. Across the 4,320 pairs of commitments at adjacent settings on the two plants, loosening ε turned a held ceiling into a breach 250 times and a breach into a held ceiling 3 times. An exact day-level sign-flip test on those pairs separates the two ends of the sweep at both targets on both plants ($p < 0.001$), and separates 11 of the 12 adjacent steps at the 5% level. The exception is the office’s smallest step, 0.05 to 0.10 at the tight target, where seven extra breaches and one reversal leave

Target	Controller	ε	Commit viol. (gap)	95% CI	vs target	Breaches	Peak kVA	Bill Rs	Solve ms
tight	marginal $q95$	—	0.106	[0.03, 0.20]	0.049	4	459.0	1,138,193	6
tight	scenario	0.05	0.097 (+0.047)	[0.04, 0.17]	0.097	9	464.6	1,141,532	52
tight	scenario	0.10	0.114 (+0.014)	[0.04, 0.21]	0.096	9	466.8	1,142,186	52
tight	scenario	0.20	0.156 (-0.044)	[0.07, 0.25]	0.073	10	469.2	1,142,647	55
tight	scenario	0.35	0.219 (-0.131)	[0.13, 0.33]	0.077	13	467.5	1,141,036	55
nominal	marginal $q95$	—	0.103	[0.03, 0.19]	0.000	0	481.9	1,149,580	6
nominal	scenario	0.05	0.078 (+0.028)	[0.02, 0.15]	0.000	0	480.5	1,148,576	63
nominal	scenario	0.10	0.097 (-0.003)	[0.03, 0.18]	0.000	0	486.0	1,151,865	63
nominal	scenario	0.20	0.150 (-0.050)	[0.07, 0.24]	0.000	0	480.1	1,148,071	63
nominal	scenario	0.35	0.208 (-0.142)	[0.12, 0.31]	0.000	0	484.0	1,150,364	55

Table 4: Closed loop over one billing month on a 13,759 m² office in Phoenix under the TNERC tariff. **Commit violation** is the acceptance metric: the fraction of horizons in which realised load cleared the ceiling the optimiser committed to, which is what ε is a statement about. The against-target column is the business metric and is not what the chance constraint promises, since the committed peak is a decision variable. Rank correlation 0.957, mean absolute gap 0.057, conservative at 5 of 8 levels; resolution floor $1/S = 0.025$. Intervals are a day-block bootstrap over the sampled commitments, which are overlapping 16-hour windows and so are not the independent draws a binomial interval would assume; the requested ε lies inside its own interval at 6 of 8 levels, and the mean half-width is 0.081, which is the resolution one billing month buys.

Target	Controller	ε	Commit viol. (gap)	95% CI	vs target	Breaches	Peak kVA	Bill Rs	Solve ms
tight	marginal $q95$	—	0.297	[0.18, 0.42]	0.101	13	890.9	2,837,240	6
tight	scenario	0.05	0.178 (+0.128)	[0.08, 0.28]	0.101	14	885.5	2,833,912	51
tight	scenario	0.10	0.233 (+0.133)	[0.13, 0.35]	0.124	15	895.3	2,835,693	61
tight	scenario	0.20	0.300 (+0.100)	[0.19, 0.42]	0.123	14	903.8	2,838,637	62
tight	scenario	0.35	0.392 (+0.042)	[0.26, 0.52]	0.075	9	915.4	2,842,287	63
nominal	marginal $q95$	—	0.297	[0.18, 0.42]	0.000	0	914.0	2,843,724	6
nominal	scenario	0.05	0.172 (+0.122)	[0.08, 0.27]	0.023	1	945.5	2,849,643	61
nominal	scenario	0.10	0.236 (+0.136)	[0.13, 0.36]	0.023	1	932.7	2,846,024	63
nominal	scenario	0.20	0.300 (+0.100)	[0.19, 0.42]	0.023	1	931.3	2,846,298	81
nominal	scenario	0.35	0.392 (+0.042)	[0.26, 0.52]	0.023	1	938.9	2,848,913	59

Table 5: Closed loop over one billing month on the IIIT-Delhi campus feed under the DERC tariff, the billed consumer of Section 4. **Commit violation** is the acceptance metric: the fraction of horizons in which realised load cleared the ceiling the optimiser committed to, which is what ε is a statement about. The against-target column is the business metric and is not what the chance constraint promises, since the committed peak is a decision variable. Rank correlation 0.988, mean absolute gap 0.100, conservative at 0 of 8 levels; resolution floor $1/S = 0.025$. Intervals are a day-block bootstrap over the sampled commitments, which are overlapping 16-hour windows and so are not the independent draws a binomial interval would assume; the requested ε lies inside its own interval at 4 of 8 levels, and the mean half-width is 0.113, which is the resolution one billing month buys.

only five days with any net difference ($p = 0.0625$). A day-block bootstrap calls that step resolved too, and we do not: when the evidence sits on a handful of days, a bootstrap that resamples only what was observed is overconfident. The same matched comparison settles a question the tables leave open. On the campus the tightest scenario setting cuts commit violation below the incumbent marginal controller’s by 0.119 and 0.125 at the two targets ($p \leq 0.001$); on the office the two cannot be told apart ($p = 0.6875$ and 0.125). So within one billing month the dial’s response is resolved step by step; what one month does not resolve is how close each level lands to the ε it was asked for.

The dial’s *calibration* does not replicate. Mean absolute gap is 0.100 against 0.057, and the sweep is conservative at *none* of its eight levels rather than five: on this feed the formulation delivers more risk than it was asked for at every setting. The intervals sharpen where that matters: they exclude the requested level at $\varepsilon = 0.05$ and 0.10 on both targets — the two tightest settings, the ones an operator would actually use — and contain it at 0.20 and 0.35, four of eight in all. We do not think this is the risk formulation failing, and the reason is one row of Table 2. The campus feed is the single window in the study whose per-step layer *under-covers* — $\hat{\alpha} = 0.079$ against a nominal 0.05, the only figure above the line — and ε is a statement built on top of that layer. A joint guarantee cannot be more exact than the marginals it composes, and here the marginals are optimistic by half again.

That is a testable claim, so we tested it. Restating the adaptive step in the interval’s units — which Section 7 shows is the defensible way to set it — moves the campus per-step rate from 0.079 to 0.069, and the whole sweep was rerun on that calibration with nothing else changed. The dial’s exactness improves in the direction the explanation predicts, and by an amount this experiment cannot separate

Series	June	realised	95% CI	ν_{tail}	copula	ν_{lik}	copula	copula, $\nu=7$
Phoenix office (BDG2)	2017	0.405	[0.28, 0.53]	7	0.497	25	0.567 [†]	0.497
IIIT-Delhi Academic	2015	0.393	[0.28, 0.50]	3	0.477	25	0.714 [†]	0.616 [†]
IIIT-Delhi Academic	2016	0.636	[0.50, 0.75]	5	0.580	25	0.732	0.631
IIIT-Delhi Academic	2017	0.422	[0.30, 0.54]	5	0.595 [†]	Gauss.	0.801 [†]	0.648 [†]
IIIT-Delhi Boys	2014	0.501	[0.38, 0.61]	4	0.554	25	0.746 [†]	0.642 [†]
IIIT-Delhi Campus	2017	0.432	[0.29, 0.58]	4	0.482	25	0.636 [†]	0.553
IIIT-Delhi Dining	2017	0.714	[0.61, 0.81]	15	0.744	25	0.774	0.665
IIIT-Delhi Facilities	2015	0.639	[0.55, 0.71]	7	0.705	Gauss.	0.875 [†]	0.705
IIIT-Delhi Facilities	2016	0.621	[0.52, 0.72]	5	0.633	Gauss.	0.856 [†]	0.690
IIIT-Delhi Facilities	2017	0.739	[0.62, 0.85]	10	0.753	Gauss.	0.876 [†]	0.706
IIIT-Delhi Girls	2014	0.693	[0.59, 0.78]	4	0.589 [†]	Gauss.	0.848 [†]	0.684
IIIT-Delhi Girls	2017	0.818	[0.74, 0.89]	7	0.739	Gauss.	0.914 [†]	0.739
IIIT-Delhi Library	2014	0.480	[0.36, 0.59]	4	0.567	Gauss.	0.816 [†]	0.659 [†]
IIIT-Delhi Library	2016	0.509	[0.41, 0.61]	4	0.504	25	0.670 [†]	0.579

Table 6: What selecting the copula’s degrees of freedom contributes. Realised horizon exceedance at $H = 64$ on the held-out month with its day-block bootstrap interval, and the copula’s prediction under three choices of ν : matched to horizon exceedance on the validation block (the paper’s choice; columns 5–6), maximising the t -copula pseudo-likelihood on the same block (columns 7–8), and held at the Phoenix value $\nu = 7$ with no refitting (last column). [†] marks a prediction outside the realised interval. The tail-matched choice lies inside on 12 of 14 windows (mean absolute error 0.065). The likelihood criterion selects $\nu \geq 25$ or the Gaussian on every window, over-predicts on 14 of 14 and lies inside on 2: it is dominated by the body of the distribution, where the families agree, and cannot see the upper-tail dependence the horizon event is made of. The Phoenix value applied unchanged to the 13 Indian windows lies inside on 9 of 13 (mean absolute error 0.098); the per-window selection buys accuracy, not the agreement itself. The independence figure lies inside on none.

from noise. Mean absolute gap falls from 0.100 to 0.083 and realised violation at $\varepsilon = 0.05$ from 0.178 to 0.158, both toward the 0.057 and 0.097 of the well-calibrated plant; rank correlation is unchanged at 0.988 and the sweep is still conservative at none of its eight levels. But each of those rates is a fraction of the commitments sampled over one month, and the day-block interval on it is wider than the movement — a mean half-width of 0.113 on the campus sweep against a shift of 0.018 — so what this run establishes is a direction and not a size. The claim the paper makes is therefore the weak one: a per-step layer that under-covers is a plausible and partially-supported cause of a dial that over-delivers risk, the repair moves the dial the right way, and the remaining distance to the calibrated plant is neither shown to close nor shown to persist. Separating those would need more billing months than one.

The honest summary of Contribution 2 is therefore narrower than one control object suggested. On both plants ε converts a risk level that was an emergent property of horizon length into one an operator sets and gets a monotone response from. On the plant whose marginals are calibrated it is also roughly exact and errs safe; on the plant whose marginals are not, it is monotone and optimistic, and repairing those marginals moves it the right way by an amount one month cannot resolve. That is a useful thing for a deployment to know, and it says the marginal calibration layer is not a preliminary to the guarantee but part of it.

What it does not do, stated in full. It is not exact. At the tightest setting, $\varepsilon = 0.05$ delivers 0.097 at the tight target, nearly twice the requested risk and on the unsafe side — though one month’s interval on it, 0.04–0.17, cannot tell that miss from none. The dial has two resolution limits. One is statistical, and it bears on exactness rather than response: one month separates the settings from one another, as shown above, but each level’s own interval is too wide to say how far it lands from its ε , and narrowing that would take more billing months, not more scenarios. The other is a hard floor at $1/S = 0.025$: with S scenarios the finest expressible risk level is one scenario’s worth of probability mass, below which the CVaR surrogate collapses onto the sample maximum and successive ε cease to be distinguishable. That is a property of sample average approximation rather than a tuning artefact, and a finer dial costs scenarios and solve time.

Two further caveats belong here rather than in a footnote, and one that was a caveat has been measured. The copula’s degrees of freedom are selected on the validation block by matching predicted to realised horizon exceedance, which is the quantity the copula is later judged on; the split is honest, but the criterion knows what it will be graded on. Table 6 bounds what that choice contributes. Holding ν at the Phoenix value of 7 and applying it unchanged to the thirteen Indian windows, the copula still lies inside the realised interval on nine of thirteen, with mean absolute error 0.098 against 0.063 when ν is selected

Site	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Phoenix, United States	United States	0.87	0.38	4.529	3.484	+0.231	0.884
Austin, United States	United States	0.75	0.13	1.352	1.061	+0.215	0.906
Washington DC, United States	United States	0.31	0.23	4.698	2.964	+0.369	0.900
Minneapolis, United States	United States	0.11	0.13	21.875	10.792	+0.507	0.916
Berkeley, United States	United States	0.05	0.26	5.855	3.033	+0.482	0.873
London, United Kingdom	United Kingdom	0.05	0.04	4.106	2.061	+0.498	0.910
Dublin, Ireland	Ireland	0.00	0.04	7.636	3.451	+0.548	0.918

Table 7: Climate arm: demographic held fixed at Education, climate and country varied across seven sites and three countries.

Site	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Phoenix, United States	United States	0.87	0.24	6.213	2.419	+0.611	0.888
Washington DC, United States	United States	0.31	0.05	49.625	34.038	+0.314	0.824
Minneapolis, United States	United States	0.11	0.24	4.784	3.009	+0.371	0.914
London, United Kingdom	United Kingdom	0.05	0.01	13.893	20.322	-0.463	0.930
Dublin, Ireland	Ireland	0.00	0.03	4.150	2.020	+0.513	0.912

Table 8: Office arm: the same ladder in a second usage class, run as a control on the climate arm. The London office is the study’s worst row and is reported unchanged.

per window: the selection buys accuracy, not the agreement. The textbook alternative — choosing ν by the t -copula pseudo-likelihood on the same validation block — selects $\nu \geq 25$ or the Gaussian on every one of the fourteen windows, over-predicts on all fourteen, and lies inside the interval on two. The likelihood is dominated by the body of the distribution, where the two families agree; the horizon event is a statement about the joint upper tail, where they do not. That is why the one free parameter is fitted to the decision-relevant statistic rather than to the likelihood, and it is a warning for anyone fitting a dependence model for a tail event by the default criterion. Base-load and PV scenarios are drawn independently, while cloud physically raises cooling load at the same moment it cuts PV output, so the true joint law has worse afternoons than these scenarios contain. And the horizon is sixteen hours while the demand charge bills a monthly maximum over roughly 2,880 intervals: a joint guarantee over the window is strictly stronger than a marginal one and strictly weaker than a monthly one. Extending it needs the copula fitted over a month-length horizon, which is a 2,880-square correlation matrix and needs structure rather than a raw estimate.

The honest summary is not that the guarantee is restored exactly. It is that a risk level which was previously an *emergent property of horizon length* — unstated, unsettable, and equal to 0.106 at the tight target on the office and 0.297 on the campus feed under the incumbent marginal controller — is now a parameter with a monotone response on both, conservative on the one whose marginals are calibrated, at a cost of Rs 3,339 on the month there and Rs 3,329 *saved* on the campus, where the tight target binds so rarely that the scenario controller buys its risk level out of headroom the marginal one was wasting.

7 What degrades with aggregation, and what does not

One protocol — 15 months train, 2 months validate, 1 month test, temporal throughout, conformal calibration on the validation block — run unchanged over thirty supplies. Tables 7–11 give the five arms.

Table 12 and the upper panel of Figure 3 are the benchmark’s own reading, and they were the third contribution of the first draft of this paper. A nominal 90% interval holds at building level (mean 0.901; one row of eighteen below 0.85) and fails at system level (mean 0.829 metered, 0.824 reconstructed; five of six and two of six below 0.85 respectively); Delhi is the worst metered row at 0.762 and Heilongjiang the worst row anywhere at 0.565. The audit of the adaptive layer at system scale, later in this section, is where that reading came apart. Every row of the benchmark was calibrated with split conformal and an adaptive layer whose step, $\gamma = 0.35$, is a number in the units of the series: on a building whose intervals are tens of kilowatts wide it is a working layer, and on a grid whose intervals are thousands of megawatts wide it is one that does not move. The building rows carried a remedy the system rows did not, and the ladder measured the remedy’s absence.

Table 13 is the same measurement without the confound: the paper’s forecaster refitted once on every supply and the same predictions calibrated three ways. Under split conformal alone the tiers sit within 0.03 of each other — 0.820 on buildings, 0.791 on metered system demand, 0.785 reconstructed — and

Usage class	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Entertainment/public assembly	United States	0.31	0.31	4.193	2.038	+0.514	0.919
Healthcare	United States	0.31	0.21	2.907	1.702	+0.415	0.902
Retail	United States	0.31	0.25	7.681	4.734	+0.384	0.897
Warehouse/storage	United States	0.31	0.06	1.635	1.189	+0.273	0.942
Lodging/residential	United States	0.31	0.28	2.726	2.102	+0.229	0.901
Public services	United States	0.31	0.03	3.882	3.680	+0.052	0.887

Table 9: Demographic arm: climate held fixed at Washington DC, demographic varied across the eight usage classes that pass the selection rule at that site. Six rows appear here; Education and Office were also run in this arm and are shown in Tables 7 and 8 instead, since a building is one supply however many arms it appears in.

Series	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
India (Delhi)	India	0.89	—	72.499	65.437	+0.097	0.762
Spain	Spain	0.28	—	594.320	226.544	+0.619	0.813
France	France	0.08	—	942.571	342.809	+0.636	0.829
Germany	Germany	0.06	—	1344.698	386.465	+0.713	0.837
United Kingdom	United Kingdom	0.02	—	451.549	378.932	+0.161	0.822
Ireland	Ireland	0.00	—	29.812	16.624	+0.442	0.911

Table 10: National arm, metered: system-level demand in six countries. Delhi is 15-minute native resolution from the Delhi State Load Despatch Centre; the European series are ENTSO-E hourly via Open Power System Data.

eleven of the eighteen buildings are below 0.85. The exchangeability failure is universal, not a property of aggregation. Under the adaptive layer with its step stated as a fraction of the interval width — $\kappa = 0.006$, the width read from the validation block so that nothing from the test month is used — every tier is at the level, 0.898, 0.914 and 0.925, and no supply in the study is below 0.85: Heilongjiang is 0.937 and Delhi 0.904. The building–system gap is +0.029, +0.073 and −0.016 under the three calibrations. What degraded with aggregation was the step.

This is consequential rather than curious, because the controller’s $q95$ is read from the same object, and the consequence has changed sign since the first draft. The hazard is not a tier the guarantee cannot reach. It is a parameter that goes quietly inert when the layer is moved to a larger meter, while the coverage it reports looks like a property of the meter.

The mechanism. Split conformal guarantees coverage when the calibration and test blocks are exchangeable [Vovk et al., 2005, Romano et al., 2019]. A Delhi May is not exchangeable with a Delhi June, because the monsoon arrives between them and changes the relationship between temperature and load. This is an assumption failure, not a tuning failure, and it is diagnosable rather than speculative: our own audit partitions May into six disjoint calibration blocks and finds coverage varying from 0.710 to 0.861 with a block-bootstrap standard error of 0.031, failing every acceptance criterion stated in advance.¹

The remedy, at building scale. Table 14 and Figure 4 run a walk-forward year. Split conformal ranges over 0.447–0.936 by month and spends 52% of the year inside a 0.85–0.95 band; adaptive conformal inference [Gibbs and Candès, 2021] ranges over 0.867–0.928 and spends 87%. Under a frozen model and a synthetic distribution shift the same ordering holds: split conformal settles at 0.770 coverage and never recovers within the run, ACI at 0.885. The layer with the finite-sample theorem is not the layer that delivers the level here, and that is worth stating plainly because it inverts the usual presentation order.

The remedy, at system scale. The same audit on the Delhi city feed — the tier where the failure is systematic, and the tier India sits in — is the second and third rows of Table 14 and the lower panel of Figure 4, the 2016–17 protocol moved five years back onto the 2011–12 archive so that the calendar position is identical. Split conformal fails harder here than at the building. Six disjoint May calibration blocks give June coverage from 0.344 to 0.728, mean 0.524 with a day-block bootstrap standard error of 0.044; the walk-forward year ranges 0.628–0.980 by month and spends 48% in band; after the frozen shift it settles at 0.358. The adaptive layer, carried over with the step it had at the building, $\gamma = 0.35$, reaches 69% in band and 0.733 after the shift: better, and not the level.

¹The bootstrap resamples thirty whole test days rather than 184,320 overlapping 15-minute forecasts, because those are not 184,320 independent observations. The naive standard error is under 0.001 and would pass any model ever built.

Series	Country	Load-temp.	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Hainan (Haikou)	China	+0.10	0.97	–	131.650	93.878	+0.287	0.960
Guangdong (Guangzhou)	China	+0.37	0.85	–	2357.241	3097.316	-0.314	0.919
Shanghai	China	+0.07	0.44	–	445.551	352.338	+0.209	0.894
Beijing	China	-0.28	0.29	–	451.459	387.453	+0.142	0.905
Yunnan (Kunming)	China	-0.35	0.14	–	251.362	197.905	+0.213	0.703
Heilongjiang (Harbin)	China	-0.47	0.12	–	148.038	191.583	-0.294	0.565

Table 11: The China arm, added after the study was complete on a constraint imposed from outside it: six provinces spanning a wider climate range than the entire European panel, in a developing economy, at one calendar year and therefore a three-month training block rather than fifteen. Mean skill is +0.040 with 2 negative rows. **That column should not be read as a statement about the method**, for the reason established in Table 17: the series is constructed, not metered, and its construction pins our model and the baseline it is scored against to the same two numbers per day. Coverage does not run through the baseline and is readable; skill is not.

Population	n	mean cov.	worst	< 0.85
Buildings (BDG2), metered	18	0.901	0.824	1
System demand, metered	6	0.829	0.762	5
System demand, reconstructed	6	0.824	0.565	2
<i>India, one city (I-BLEND + Delhi SLDC), windows</i>				
buildings, native 15-min	12	0.910	0.897	0
campus feed (one HT consumer)	1	0.873	0.873	0
city (Delhi SLDC)	1	0.762	0.762	1

Table 12: Empirical coverage of the nominal 90% interval after conformal calibration, *as the benchmark calibrated it*: split conformal plus an adaptive layer at the step $\gamma = 0.35$ in the units of each series. Read this way the guarantee very nearly holds on individual buildings and fails systematically on system-level demand. The third row is the replication: six Chinese provinces, a different country, a different year and a different data-generating process, reproduce the system-level failure to within 0.005 of the metered panel and contain the worst row in the study (Heilongjiang, 0.565). The lower panel is the same effect inside one city: building, campus and city demand in Delhi, one weather feed, native 15-minute resolution throughout, and coverage falls monotonically with each step up the ladder. Table 13 re-reads every row of this table with the adaptive step stated in the interval’s units, and the ladder does not survive it.

But γ is a number in the units of the series. On a building whose intervals are 58 kW wide it moves the bound by 0.6% of the interval per step; on a city whose intervals are 739 MW wide, by 0.05%. The same setting is a layer twelve times slower, and Table 15 shows that this is the whole of the difference: ACI replayed over the saved year at a grid of steps stated as a fraction κ of the interval width, nothing retrained, reaches the level at the city as it does at the building. Carried over as the building’s $\kappa = 0.006$ rather than its γ — nothing tuned on Delhi — the adaptive layer spends 100% of the year in band with by-month coverage 0.874–0.935, holds the monsoon June that the benchmark row reports at 0.762 at 0.895, and settles at 0.887 after the shift, where the building settled at 0.885. The level is restored at system scale. What did not transfer was the parameter’s units, and a practitioner moving the layer from a building to a feeder would hit exactly this. One caveat is due: above $\kappa \approx 0.01$ the interval is a fast tracker rather than a forecast interval, since the recursion pins the long-run exceedance rate whatever the model and the width it reports is no longer the forecaster’s; the sweep is read for the transfer, not for the numbers at its top.

Selection is regional, and we found that out the hard way. The rule’s final clause — positive load-temperature correlation — is a sanity check in Phoenix, where every candidate is cooling-dominated, and a site-destroying filter in Minneapolis or Dublin, where buildings consume electricity when it is cold. Seventy-seven buildings across nine sites pass every clause but that one. Separately, the two hottest sites in the corpus, the closest available analogues to an Indian metro climate, yield no admissible building at all for want of coverage in the split window. Both facts were discovered by applying the rule, not anticipated when writing it.

Population	n	κ of $\gamma=0.35$	split	ACI $\gamma=0.35$	ACI $\kappa=0.006$	worst	< 0.85
Buildings (BDG2), metered	18	0.0113	0.820	0.901	0.898	0.862	0
System demand, metered	6	0.0001	0.791	0.829	0.914	0.902	0
System demand, reconstructed	6	0.0001	0.785	0.824	0.925	0.902	0
<i>India, one city, windows</i>							
buildings, native 15-min	12	0.0338	0.868	0.910	0.908	0.874	0
campus feed (one HT consumer)	1	0.0016	0.819	0.873	0.888	0.888	0
city (Delhi SLDC)	1	0.0006	0.605	0.762	0.904	0.904	0

Table 13: The ladder of Table 12 re-read without the confound. The paper’s forecaster refitted once per supply and the same predictions calibrated three ways: split conformal alone; split plus ACI at the absolute step $\gamma = 0.35$ every benchmark row used; and split plus ACI at the building audit’s step as a fraction of the interval width, $\kappa = 0.006$, the width taken from the validation block so that nothing from the test month is used. The third column is what the absolute step amounted to on each population, as a fraction of its interval (median): a working adaptive layer on buildings and a nearly inert one on system demand. Mean coverage of the nominal 0.90 interval on the test month, then the worst row and the count below 0.85 under the relative step. The building–system gap is +0.029 under split conformal alone, +0.073 under the absolute step the benchmark used, and -0.016 under the step stated in the interval’s units.

Series	γ	κ	split, by month	in band	ACI, by month	in band	split	ACI
Phoenix office, building, 2016–17	0.35	0.0060	0.447–0.936	52%	0.867–0.928	87%	0.770	0.885
Delhi city, system, 2011–12	0.35	0.0005	0.628–0.980	48%	0.765–0.969	69%	0.358	0.733
Delhi city, system, 2011–12	4.43	0.0060	0.628–0.980	48%	0.874–0.935	100%	0.358	0.887

Table 14: The adaptive layer at each tier it has been run at. Walk-forward year of twelve monthly folds, each trained strictly on the past and calibrated on the thirty days before it; coverage of the nominal 0.90 interval by month (range) and the share of the rolling 30-day curve inside 0.85–0.95. The last two columns are coverage after a level and volatility shift injected into a model frozen six months earlier, which static calibration cannot respond to by construction. γ is the ACI step in the units of the series and κ the same step as a fraction of the mean split-conformal interval width at the audited lead. At the building, split conformal spends 52% of the year in band and ACI 87%. Carried to the city as the same absolute γ , the step is 13 $\times$ smaller relative to the interval and ACI reaches 69% against split’s 48%. Carried as the same κ , with nothing tuned on Delhi, ACI spends 100% of the year in band, by-month coverage 0.874–0.935, and settles at 0.887 after the shift against split’s 0.358.

8 Skill and flexibility are independent axes

We expected forecast skill — measured with pinball loss, a strictly proper scoring rule [Gneiting and Raftery, 2007], against each series’ own seasonal-naïve baseline — to increase with cooling load, on the reasoning that weather-driven demand is the learnable part. Table 16 and Figure 5 report the opposite: a null. Every correlation against skill is indistinguishable from zero and every leave-one-out range straddles or nearly straddles the sign line, which at these sample sizes is what a null result looks like. It replicates at $r = +0.054$ on the Chinese provincial panel, which was added afterwards and spans a wider climate range than the whole European set.

The controllable fraction, by contrast, tracks climate strongly ($r = +0.489$), running from 37.6% of the meter in Phoenix to 0.8% in London.

The deployment consequence is a screening rule. Do not ask whether a site’s load is predictable; ask how much of it is weather-driven. A Dublin office is highly predictable (skill +0.513) with roughly 3% controllable load, and there is nothing to sell there. Conflating the axes inverts the decision. For the target market this is favourable: Delhi demand correlates +0.59 with outdoor temperature against -0.64 for France, placing India at the useful end of the axis that matters.

9 The new country, and what it turned out to measure

The study was complete when a constraint was imposed from outside it: add a country. China was chosen because six provinces spanning Hainan to Heilongjiang cover a wider climate range than the entire European panel and do so in a developing economy, which is the comparison India actually needs. The protocol adaptation it forced is stated in Table 11 and is a single change: the source is one calendar year, so the fifteen-month training block does not exist and the calendar *position* of the split is preserved instead — validate on April and May, test on June, as in every other row — leaving three months of

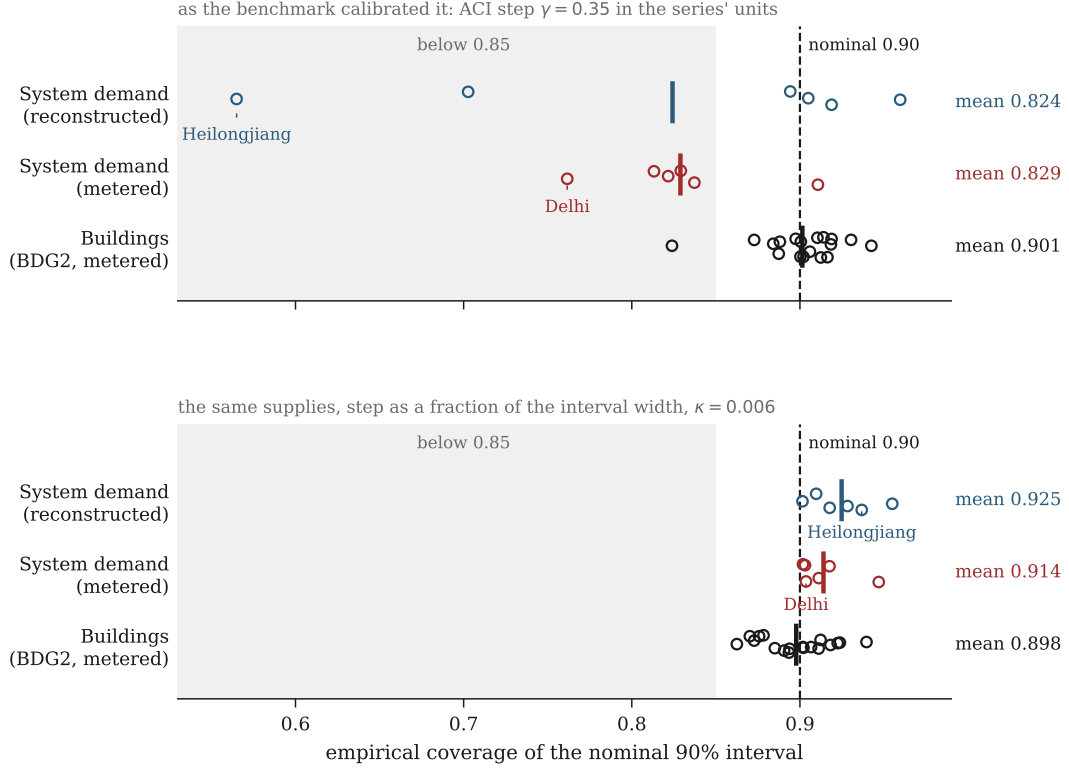


Figure 3: One dot per supply, twice. Top, as the benchmark calibrated it: the nominal 90% interval very nearly holds on individual buildings and fails once demand is aggregated to system level, on the metered panel and the reconstructed one alike, two populations sharing no country, no year and no data-generating process whose means differ by 0.005. Bottom, the same supplies with the adaptive layer’s step stated as a fraction of the interval width rather than as a number in the series’ units: every tier is at the level and the ladder is gone.

training rather than fifteen. That biases against us, which is the safe direction to err.

The audit. This is the only tier-2 source whose documentation describes a construction rather than a measurement. Wu and Kan take each day’s maximum and minimum from National Development and Reform Commission reporting, trace a representative workday profile and a representative holiday profile off published load curves with WebPlotDigitizer, and rescale one of those two profiles to each day’s envelope [Wu and Kan, 2023]. Taking that on trust is exactly what this project does not do elsewhere, so we checked it against the data: min–max normalise every calendar day’s 24-hour profile and count the distinct shapes in a year.

The answer is two, in every province. Table 17 reports it: 778 numbers — two normalised day-shapes and 365 (min, max) pairs — reproduce all 8,760 hourly values to within 2×10^{-9} of their magnitude, which is the precision the source is published at rather than an approximation we introduced, at a compression of better than eleven to one. The documentation is accurate.

What that disqualifies. Skill in this study is a ratio against each series’ own seasonal-naïve baseline. When every workday shares one shape, seasonal naïve recovers that shape exactly and its entire error is the daily envelope — and so is ours. Both models collapse onto predicting the same two administrative numbers per day, the ratio between them collapses toward zero, and the observed mean skill of +0.040 with two negative rows is what that predicts rather than a finding about a forecaster. The China skill column is reported for completeness and is not evidence either way about the method.

This costs us the arm’s most quotable result. Guangdong is the closest structural analogue to Delhi anywhere in the study — cooling share 0.85 against Delhi’s 0.89, load–temperature correlation +0.37 against +0.59, a comparably peaky profile — and it returns -0.314 , the second-worst skill in the benchmark. Read naively that corroborates the Indian result on independent 2018 data from another country and retires the objection that Delhi’s weakness is an artefact of a decade-old archive. It does not, because the number is a property of the construction. We report the temptation as well as the retraction, because the inference is one a reader would otherwise make on our behalf.

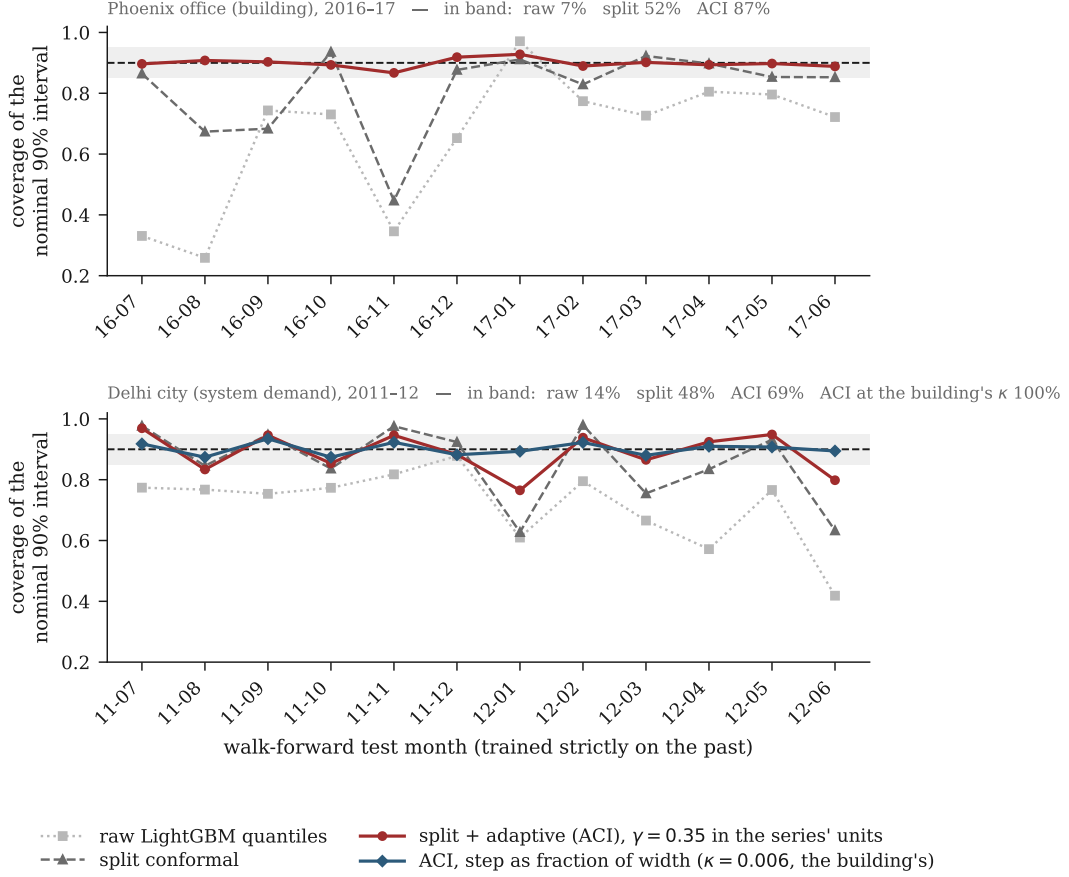


Figure 4: The remedy, measured at both tiers. Twelve monthly folds, each trained strictly on its own past and calibrated on the thirty days immediately before it. Top, the Phoenix office: raw quantiles are not calibrated out of sample at all; split conformal — the layer with the theorem attached — narrows the failure and does not remove it, bottoming out at 0.447 in a month whose ceiling was being defended against a bound reality broke through a third of the time; adaptive conformal inference holds 0.867–0.928 in every month. Bottom, the Delhi city feed under the identical protocol moved five years back onto its archive. Red is ACI at the building's step $\gamma = 0.35$, a number in the series' units; blue is the same step carried over as the building's fraction of the interval width, $\kappa = 0.006$, with nothing tuned on Delhi.

What survives. Coverage is a property of interval width against realised error and does not run through the baseline, so the tier-2 result can be read on this arm — and it replicates twice over. Under the benchmark's calibration the six provinces average 0.824 against 0.829 on the metered panel, on an error process unlike any other in the study, with the worst row anywhere at 0.565; under the adaptive step stated in the interval's units they average 0.925 against 0.914, and the worst row, Heilongjiang, is 0.937 (Table 13). Two populations sharing no country, no year and no data-generating process landing within 0.005 of each other under the inert layer and within 0.011 under the working one is stronger evidence for both readings than a larger metered sample would have been: the failure is universal, and so is the remedy. The null of Section 8 replicates too. The wider coverage spread on this panel under the benchmark's calibration (0.565–0.960 against 0.762–0.911) is consistent with the three-month calibration history and is not separately identified.

What the exercise is worth. A constraint imposed late produced no new capability and one new measurement, which is a characterisation of the dataset rather than of the method, and which removed a result we would have liked to keep. That is the outcome we would want from an audit, and it is reported at the same size as everything else.

κ	in band	by month	width	in band	by month	width
0 (split only)	52.2%	0.447–0.936	58	48.1%	0.628–0.980	739
0.001	61.4%	0.785–1.000	67	72.9%	0.825–0.960	756
0.002	81.8%	0.816–0.992	64	89.7%	0.840–0.956	735
0.005	85.2%	0.860–0.941	61	99.6%	0.869–0.939	698
0.01	93.1%	0.878–0.932	59	100.0%	0.882–0.921	672
0.02	100.0%	0.890–0.924	53	100.0%	0.894–0.908	629
0.05	100.0%	0.896–0.909	46	100.0%	0.897–0.904	560
0.1	100.0%	0.897–0.906	41	100.0%	0.898–0.902	495

Table 15: ACI replayed over the saved walk-forward year at a grid of step sizes, stated as a fraction κ of the mean split-conformal interval width at the audited lead (58, 739 in the series’ units for the Phoenix office and Delhi city respectively; left block Phoenix, right block Delhi). Nothing is retrained: ACI is a pass over the saved split-conformal predictions in time order. In band is the share of the rolling 30-day coverage curve inside 0.85–0.95; by month is the range of monthly coverage; width is the mean adaptive interval. The audit’s absolute default $\gamma = 0.35$ is $\kappa = 0.0060, 0.0005$ on the two series. Above $\kappa \approx 0.01$ the interval is a fast tracker rather than a forecast interval: the recursion pins the long-run rate whatever the model, and the width it reports is no longer the forecaster’s.

Correlation against forecast skill	r	n	leave-one-out range
<i>Panel A — 18 BDG2 buildings, three countries</i>			
cooling-degree-day share	+0.010	18	-0.308 to +0.077
controllable (HVAC) fraction	+0.283	18	+0.029 to +0.378
mean outdoor temperature	-0.012	18	-0.266 to +0.051
median load	-0.014	18	-0.189 to +0.120
<i>Panel B — 6 Chinese provinces, independent replication</i>			
cooling-degree-day share	+0.054	6	-0.377 to +0.582
mean outdoor temperature	+0.216	6	-0.290 to +0.782
load-temperature correlation	-0.060	6	-0.619 to +0.713
<i>controllable fraction vs climate (not vs skill)</i>	+0.489	18	–

Table 16: Forecast skill is uncorrelated with climate, controllable fraction, temperature and load size, and every leave-one-out range straddles or nearly straddles zero — which at these sample sizes is what a null looks like. Panel B is the replication on a panel the null was not fitted to. The final row is the one relationship in the study that is real, and note what it is between: available flexibility tracks climate strongly, while predictability does not track anything. Skill and flexibility are independent deployment axes.

10 India in comparison

Table 18 is the comparison the benchmark exists to make, and it resolves into a single sentence: **India is the place this method is most worth deploying and the place it is currently least trustworthy.**

Why India is the favourable case. Delhi has the highest cooling-degree-day share of any metered series in the benchmark at 0.89, and it is the only series whose demand rises *strongly* with temperature (+0.59). France is its mirror at -0.64: French demand rises when it gets cold, because the building stock is electrically heated. That distinction is the whole economic argument. A controller that buys down a cooling peak inside a demand-charge tariff has something to work with in the first regime and almost nothing in the second. Delhi is also among the peakiest profiles in the set ($p99$ /median 1.61), which is precisely the shape a demand charge punishes.

Why India is the difficult case. On both axes that determine whether the controller can be relied upon, India is last among metered supplies. Skill over a seasonal-naive baseline is +0.097, against +0.713 for Germany, +0.636 for France and +0.619 for Spain — a model that needs no training at all is nearly as good on Delhi as ours. And coverage of the nominal 90% interval is 0.762 under the benchmark’s calibration, the worst metered figure anywhere in the study against 0.911 for Ireland — and 0.904 once the adaptive step is stated in the interval’s units (Table 13), which is the reading that stands.

The second of these is the serious one. The controller’s $q95$ is read from the same object as that interval, so an interval that is too narrow yields a demand ceiling that is too narrow, and the safety

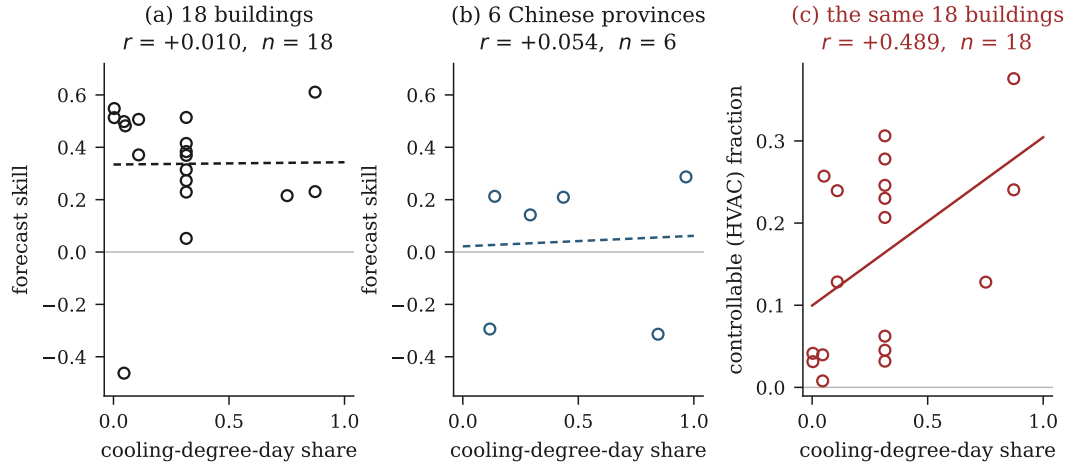


Figure 5: (a) The null. (b) The same null on a panel it was not fitted to, in a different country, year and data-generating process; panels (a) and (b) share a vertical scale. (c) The same eighteen buildings, the same horizontal axis, and the one relationship in the study that is real. How much load there is to shift tracks climate. How well it can be predicted does not.

Province	Days	Shapes	Days/shape	Numbers	Values	Max err. %
Hainan (Haikou)	365	2	115/250	778	8,760	0.000000
Guangdong (Guangzhou)	365	2	115/250	778	8,760	0.000000
Shanghai	365	2	115/250	778	8,760	0.000000
Beijing	365	2	115/250	778	8,760	0.000000
Yunnan (Kunming)	365	2	115/250	778	8,760	0.000000
Heilongjiang (Harbin)	365	2	115/250	778	8,760	0.000000

Table 17: Provenance audit of the China arm, run against the data rather than taken from its documentation. Every calendar day’s 24-hour profile is min-max normalised and the distinct shapes counted. *Shapes* is how many survive; *Numbers* is how many values are needed to rebuild the year from them; *Values* is how many the year contains. A whole year of hourly provincial demand is 778 numbers — 2 normalised day-shapes and one (min, max) pair per day — reproducing all 8,760 values to within 2e-09 of their magnitude, which is the precision the source is published at rather than an approximation we introduced. Compression 11.26:1. The published method is exactly what the authors state it is. The consequence for this study is that forecast skill, a ratio against a seasonal-naive baseline, is uninformative on these series: the shape the baseline has to guess is constant, so both models reduce to predicting the same two administrative numbers per day.

property is weakest on exactly the load the project targets.

At building level, the verdict is the same. Table 19 is the Indian building arm: the IIIT-Delhi campus, thirteen windows, the unchanged protocol. The skill column says what the city-level row said. Over the twelve building windows the model beats a seasonal-naive baseline on six and loses on six, with mean skill -0.037 ; the worst row in the entire study is now an Indian one (Library, June 2016, -0.965 , on a seven-month training block). Skill tracks the length of the training block — $+0.060$ on the four windows with thirteen months or more, -0.086 on the eight shorter ones — which is the direction the China arm’s adaptation was argued to err in, now measured. The campus feed, the one object here that a demand charge is actually levied on, returns $+0.083$: a model that needs no training is nearly as good. That is India’s skill verdict, and it no longer rests on a decade-old city archive.

The calibration column says something the city-level row could not, and it says it twice. Under the benchmark’s calibration coverage holds at building scale, 0.897 – 0.941 on every window, exactly as it did on the eighteen BDG2 buildings; it is 0.873 on the campus feed, and 0.762 on the city. Table 12 carries this as its lower panel: one city, one weather feed, three levels of aggregation, and coverage falling with each step up the ladder — which reproduced the first draft’s aggregation effect inside a single city with nothing else varying, and reproduced its confound with it. Under the adaptive step stated in the interval’s units the same three levels read 0.908 , 0.888 and 0.904 (Table 13): flat within 0.02 , in a city where the step the building used was one sixth of the building’s on the campus feed and one twentieth on the city. The campus row is the one that matters for control: the per-step exceedance of its $q95$ is 0.079

Series	Provenance	Native min	Cooling share	Load-temp. corr.	$p99/med$	Skill	Cov	90%
Hainan (Haikou)	recon.	60	0.97	+0.10	1.32	+0.287	0.960	
India (Delhi)	metered	15	0.89	+0.59	1.61	+0.097	0.762	
Guangdong (Guangzhou)	recon.	60	0.85	+0.37	1.47	-0.314	0.919	
Shanghai	recon.	60	0.44	+0.07	1.64	+0.209	0.894	
Beijing	recon.	60	0.29	-0.28	1.59	+0.142	0.905	
Spain	metered	60	0.28	+0.17	1.31	+0.619	0.813	
Yunnan (Kunming)	recon.	60	0.14	-0.35	1.24	+0.213	0.703	
Heilongjiang (Harbin)	recon.	60	0.12	-0.47	1.31	-0.294	0.565	
France	metered	60	0.08	-0.64	1.58	+0.636	0.829	
Germany	metered	60	0.06	-0.10	1.32	+0.713	0.837	
United Kingdom	metered	60	0.02	-0.13	1.40	+0.161	0.822	
Ireland	metered	60	0.00	-0.07	1.37	+0.442	0.911	

Table 18: India against every other system-level series in the study, ordered by cooling-degree-day share. Among metered supplies Delhi is the extreme of both halves of the verdict: the highest cooling share, the strongest positive load-temperature relationship, the peakiest profile, and simultaneously the lowest forecast skill and the worst interval coverage. The six Chinese provinces, added after the study was written, displace India from three of those four extremes on the face of the table — and the provenance column is why that displacement is reported rather than acted on. Guangdong is the closest structural analogue to Delhi in the study and returns the second-worst skill in it, which reads like corroboration and is not: Section 9 shows the Chinese skill column is an artefact of how the data was built. Delhi remains the only series at native 15-minute resolution, the cadence the controller runs at and the cadence Indian demand charges are assessed on.

against a nominal 0.05 under the benchmark’s calibration, the only window in Table 2 that is over the line. It is also the one window in that table where the absolute step was *slower* than the relative one rather than faster ($\kappa = 0.0015$ against 0.006), which is the campus feed being the largest Indian meter in the study; restated in the interval’s units its per-step rate falls to 0.069 and its horizon exceedance is unchanged at 0.438 (Table 3).

What this arm cannot say. Six of the seven buildings are on a central chiller that is not on their meter, so the building rows are forecasting and calibration results and no savings figure is claimed from them. The campus feed is a control object — it carries the chiller, and it is the consumer the tariff bills — and Section 6 runs the closed loop on it under Delhi’s own tariff, which closes the loop on an Indian consumer at the resolution and under the tariff the bill is actually computed at. What that run found, and did not find, is reported there.

11 What India must fix

Stated as work rather than as caveats, in descending order of how much they cost.

1. No row of this study is both Indian and building-level. The Building Data Genome Project 2 harmonises 1,636 buildings across three countries and contains no Indian building, so the *controller* was tested on American buildings under an Indian tariff and the *forecaster* on Indian city demand, and no single row here is both. Relabelling an American building as Indian would have been worse than the gap, so the gap is reported. An earlier draft stated that no Indian building-level load dataset exists; that was wrong. I-BLEND [Rashid et al., 2019] provides 52 months (August 2013 to December 2017) of one-minute mains power and power factor for seven buildings on the IIIT-Delhi campus — academic, lecture, library, facilities, dining and two dormitories — under a CC0 licence. Three properties make it the right corpus to close this gap and not merely a convenient one. It is in Delhi, the city whose system demand is the study’s focal series. It is sampled at one minute, so the 15- and 30-minute billing blocks Indian demand charges are assessed on can be formed by averaging rather than, as for every BDG2 building here, interpolated up from hourly data that has already smoothed the peak. And it spans four Junes, so the horizon-gap measurement of Section 5 can be replicated across years on one building rather than reported once. What it lacks is submetering of HVAC, so the base-load split remains a changepoint inference. The unchanged forecasting protocol has now been run on it (Table 19); the closed loop under the Delhi (DERC) tariff has not, and nothing in this paper anticipates its result.

Series	Level	June	Train mo.	Cooling on meter	Median kW	Seas. naive	Ours	Skill	Cov 90%
Campus	campus	2017	8	yes	170.1	22.942	21.047	+0.083	0.873
Academic	building	2015	13	no	18.9	0.943	1.211	-0.284	0.903
Academic	building	2016	15	no	18.9	1.097	1.146	-0.045	0.910
Academic	building	2017	15	no	18.9	1.264	0.853	+0.325	0.909
Boys	building	2014	7	no	27.8	0.598	0.667	-0.114	0.915
Dining	building	2017	9	no	16.3	1.476	1.323	+0.103	0.901
Facilities	building	2015	10	yes	8.1	0.427	0.359	+0.160	0.902
Facilities	building	2016	5	yes	8.1	0.472	0.473	-0.001	0.897
Facilities	building	2017	15	yes	8.1	0.457	0.345	+0.246	0.903
Girls	building	2014	6	no	12.7	0.355	0.350	+0.016	0.905
Girls	building	2017	8	no	12.7	0.305	0.267	+0.124	0.908
Library	building	2014	7	no	6.2	0.546	0.553	-0.012	0.922
Library	building	2016	7	no	6.2	0.391	0.769	-0.965	0.941

Table 19: The Indian building arm: I-BLEND, IIIT-Delhi, one row per admissible (series, June) window, native 15-minute blocks, the unchanged protocol with the training block shortened where meter uptime forces it (the China-arm precedent). *Cooling on meter* is the selection rule’s chilled-water clause: six of seven buildings are served by a central chiller on the campus feed, so only the Facilities building and the campus total are control objects; every row is a forecasting and calibration result. Over the 12 building rows the model beats seasonal naive on 6; mean skill is -0.037 , $+0.060$ on the 4 windows with a full training block and -0.086 on the 8 shorter ones. Coverage is 0.897 – 0.941 on every building row and 0.873 on the campus feed.

2. Calibration collapses across the monsoon. Coverage of 0.762 is an assumption failure, not a tuning failure: split conformal requires calibration and test blocks to be exchangeable [Vovk et al., 2005, Romano et al., 2019], and a Delhi May is not exchangeable with a Delhi June. Section 7 measures the remedy on this feed: across a walk-forward year the adaptive layer, with its step carried over from the building as a fraction of the interval width, holds 0.874 – 0.935 by month where split conformal falls to 0.628 , and holds this June at 0.895 ; on the benchmark’s own June, refitted and recalibrated (Table 13), it is 0.904 . The 0.762 of Table 12 is the benchmark row under the step the building used, and it is a units artefact rather than a monsoon one: the monsoon breaks exchangeability, which split conformal cannot survive anywhere, and the layer that survives it had been switched off by its units.

3. Indian load carries calendar structure the model does not encode. A large share of Indian public holidays are lunisolar — Diwali, Holi, Eid — and move against the Gregorian calendar, so they cannot be expressed as the fixed date list that load-forecasting pipelines conventionally use, including ours until this study forced the change. Beyond holidays, Indian demand carries load-shedding events, agricultural pumping schedules and festival periods that have no counterpart in the European series and no representation in our feature set. The low skill figure of $+0.097$ should be read partly as a statement about missing features rather than solely about the data.

4. The Indian series is a decade old. The Delhi archive covers 2011–2012. India’s generation mix, air-conditioning penetration and peak demand have all moved substantially since. The load *shape* conclusions are the durable ones; the levels are not, and nothing in this paper should be read as a statement about the present Indian grid. We had a candidate answer to this objection in the China arm and Section 9 withdrew it.

What this does not undermine. The favourable half of Table 18 rests on physical structure — cooling share, the sign of the load–temperature relationship, peakiness — which is the part least likely to have inverted since 2012. Indian demand has not stopped rising with temperature.

12 Results that argue against the method

Five rows are reported unchanged, and three of them are outright negative — worse than a baseline that needs no training at all.

The London office returns skill -0.463 and public services in Washington DC returns $+0.052$. Both sit at 1–3% HVAC share, where a weather-aware model has little to add over “last week, same time” — a coherent and testable explanation rather than an excuse. Delhi returns $+0.097$ with the study’s worst metered calibration, and its explanation is different and given in Section 11: a system-level series carrying

calendar structure the feature set does not encode. Guangdong and Heilongjiang return -0.314 and -0.294 , the two lowest figures in the study, and Section 9 explains why those two should not be used against the method any more than they should have been used for it.

Two defects in our own pipeline were exposed by running it outside its home site. A hard-coded United States federal holiday calendar would have inverted the holiday flag on two of the largest annual load anomalies for every European building; holidays are now resolved per country, and India is the case that forces it, its calendar being substantially lunisolar and therefore not expressible as a fixed list. And a data-preparation stage overwrote its own manifest, silently discarding every site but the last, which single-site work had never exposed.

A third is worth recording because it is the kind that survives review. The column commonly written as “the Boole bound” in applied chance-constraint work is usually $1 - (1 - \alpha)^H$, which is the independence calculation and is not a bound. Our own tables carried that label until this draft. The corrected reading in Section 5 is stronger than the one it replaces, because the valid bound turns out to be vacuous over the horizon that matters.

13 Limitations

The system tier is a forecasting benchmark, not a control benchmark: there is no thermal mass to shift at grid level and no savings claim is made from it. The Delhi window (2011–12) does not overlap the European one (2016–17) or the Chinese one (2018), which motivates reporting within-series skill rather than absolute loss but does not eliminate the concern. Capital-city temperature proxies national load-weighted temperature, acceptable for Great Britain and Ireland and rough for Germany, France, Spain and every Chinese province. Nothing here claims that model *weights* transfer across countries — our own cold-start study finds the model losing to seasonal naive on an unseen building at the same site, scoring 3.546 pinball against that baseline’s 3.037 — only that the architecture and protocol do. The demographic arm is a single city and cannot separate demographic from local building stock. Delhi is one Indian city, and is real Indian data rather than a claim about India. The China arm is reconstructed data and Section 9 states exactly which of its columns may be read. The acceptance result of Section 6 is two plants over one billing month each; those are the deepest measurements in the study and the narrowest, and the two do not agree on the property that matters most, which Section 6 reports rather than averages.

The closed-loop results are produced on an RC plant model whose parameters are fitted from each meter’s own changepoint, but which holds a comfort band the real building evidently does not. Simulated uncontrolled peak exceeds the metered June peak by 9% on the Phoenix office and 24% on the Delhi campus: the model air-conditions continuously where the real premises let indoor temperature drift. Absolute savings from it are therefore an upper bound, and we do not claim them as metered outcomes. The acceptance result is unaffected in kind, because it grades the controller against the ceiling it committed to rather than against a kilowatt figure, but a reader should take the rupee numbers as the price of a risk level on this plant and not as a realised bill.

The five arms were run once, with the adaptive step that Section 7 then found to be stated in the wrong units, and they have deliberately not been rerun. Tables 7–11 and Table 12 are therefore reported as what that protocol produced, and Table 13 — the same forecaster, the same supplies, the same predictions, calibrated three ways — is the correction. We prefer this to a silent regeneration for two reasons. A study that quietly reruns itself after finding its own confound leaves a reader unable to see either the confound or its size; and the pinball and skill columns of the five arms do not depend on the question at issue, because every forecaster in a row passes through the identical calibration layer, so the comparison between forecasters is unaffected by which step that layer used. What the step changes is the coverage column, and that column is re-reported in Table 13 rather than overwritten.

14 Conclusion

We built an instrument rather than an algorithm and ran it unchanged over thirty electricity supplies in eight countries to find out where a tariff-native demand controller’s guarantees actually hold.

The central result is that the guarantee most commonly deployed is not the guarantee most commonly claimed. A per-step chance constraint on a demand ceiling, marginally calibrated and behaving exactly as specified, is under-protected over a sixteen-hour planning window by a factor of eight on the building where it was first measured, and by between three and sixteen across fourteen windows in two countries

under either way of tuning the layer beneath it; the figure usually quoted as the conservative correction is not a bound and overstates the risk by $2.4\times$; and the correction that is a bound is vacuous four hours in. That risk level can be turned into an operator-set parameter, at a measured price of Rs 3,339 on a billing month and with its inexactness and resolution floor stated — and on a second plant, an Indian campus feed under Delhi’s own tariff, the response stays monotone while the exactness does not survive, on the one series in the study whose per-step layer under-covers. A joint guarantee is no more exact than the marginals it composes, which makes the calibration layer part of the guarantee rather than a preliminary to it. Alongside it, a finding this paper first made and then unmade: distribution-free coverage does not degrade with aggregation level. Split conformal fails at every tier alike, and the adaptive layer holds the level at every tier once its step is stated in the interval’s units rather than the series’ — the ladder the first draft reported was that parameter going inert as the meters grew. And forecast skill and available flexibility are independent axes, only one of which tracks climate.

For India the benchmark returns a divided verdict, and both halves matter. India has the most favourable load physics in the study — the highest cooling share, the only strongly positive load–temperature relationship, a peaky profile that a demand-charge tariff punishes and this method is built to defend. It also returns the lowest metered forecast skill and the worst metered calibration we measured, and the loop has not yet been closed on an Indian building. The first half is the case for the work. The second half is the work that remains, and Section 11 states it as such rather than as a footnote — together with the Indian building corpus on which it can now be done.

Every number here is regenerated by one script from a clean checkout. The results that contradict us are printed at the same size as the rest, and the one result we most wanted to keep is the one Section 9 withdraws.

Reproducibility

Every number in this paper is written by a script from a file in `results/`; none is typed into the text. From a clean checkout the whole study regenerates in the order below: cache the system-level series and I-BLEND, run the protocol over the thirty-seven supplies, then the Indian arm window by window — once with the adaptive step as an absolute number and once as a fraction of interval width — then the audits of the China arm, the horizon bracket and closed loop, the calibration year, the step sweep, the aggregation ladder under three calibrations, the copula’s one free parameter, the figures the prose states, and the references against their index. The last two commands emit every table and figure. Each stage is resumable and skips work whose output already exists, so an interrupted run continues rather than restarts.

```
python data/national.py
python data/iblend.py
python eval/comparative.py
./reproduce_india.sh
./reproduce_india_kappa.sh
python eval/china_audit.py
python eval/horizon_risk.py
python eval/conformal_audit.py
python eval/aci_gamma.py
python eval/aci_step_check.py
python eval/copula_df_check.py
python eval/paper_numbers.py
python eval/bibcheck.py
python eval/paper_tables.py
python eval/paper_figures.py
```

The calibration audit is run a second and third time on the Delhi system series, with the protocol’s own dates shifted five years onto that series’ archive, and the adaptive step stated first as the absolute number the benchmark used and then as a fraction of interval width:

```
python eval/conformal_audit.py \
  --building IN_Delhi --shift-years 5
python eval/conformal_audit.py \
  --building IN_Delhi --shift-years 5 \
  --reuse --gamma-rel 0.006 --tag @rel
```

The second control object, the IIIT-Delhi campus feed, is billed under the DERC schedule with no photovoltaics. Its demand target is found first, and the closed loop is then run once at each adaptive step:

```
python eval/find_target.py \
    --building IIITD_Campus --tag @2017 \
    --tariff tariff/orders/derc_2021.json \
    --pv-kwp 0
python eval/horizon_risk.py \
    --building IIITD_Campus --tag @2017 \
    --tariff tariff/orders/derc_2021.json \
    --pv-kwp 0
python eval/horizon_risk.py \
    --building IIITD_Campus --tag @2017k \
    --tariff tariff/orders/derc_2021.json \
    --pv-kwp 0
```

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